

NinjaNIRS 2026/27 Upgrade Notes

Github repo for all design files that have changed compared to the public version on OpenfNIRS: https://github.com/eshazen/ninjanirs-sipm/tree/main/NN26_27/

NinjaNIRS 2026

NN26 is a planned upgrade of the two NN24 whole-head, high-density systems at BU with **high-power source optodes**, as designed and tested for the UG3 grant milestones.

Source Optodes

NIRx is in the process of manufacturing high-power source optodes for us. Assembly instructions as well as the PCB design files for the small board inside each optode are in the Github repo.

Source Splitter Boards

In testing we have used the NIRx source splitter PCBs and didn't see any issues. It might still be worth checking the current handling capacities of the PCB traces.

The traces on Eric's source splitter were designed with the higher currents in mind, but we have only manufactured a few of them.

Source Interface Cards

The NN24 VCCS Source Interface cards from NN24 can also be used for NN26, however several components need to be changed, so it is probably more economical to order new sets of PCBs. Obviously regular NIRx source optodes should not be connected to these modified boards as the LEDs could easily be overdriven!

Here are the components that need to be changed:

Designator	NN24	NN26	Notes
R9	10 Ohm	1 Ohm	Change in damping network
R13	DNP	2.2 Ohm	Change in damping network
C12	DNP	0.22 uF	Change in damping network
R8	8.2 Ohm	1.5 Ohm	Current shunt for higher amps
R30	28 kOhm	59 kOhm	Increase LED drive voltage for series cfg
F1	0ZCH0035FF2G	0ZCH0110AF2E	Increase fuse amps
R2	10 kOhm	12 kOhm	Tune max amps slightly lower

Backplane

The NN24 backplane should be fine. Any backplane with “VCCS” in the designator should be compatible. Only the oldest (v1) backplanes are not compatible as they were designed for the first version source cards without current source.

Control Card

Currently one NN24 system has a v2 control card, the other one has a v3 control card. Both are compatible with the high power optodes, though it would make sense to upgrade the system with the v2 card to v3 when working on it. V3 supports the voltage/current monitoring Sreekanth added to the NinjaWeb.

System Case

The current case should be fine.

Firmware / NinjaWeb GUI

It would be highly advisable to implement some additional safety features to prevent the average optical power of a single optode to exceed ANSI limits. This could currently happen if someone e.g. loads a wrong probe file for much smaller probe so that now the duty cycle of a single optode is much higher.

This safety could e.g. be implemented in the source card firmware, running separately on the second core of the MCU which is currently not used. Or a simpler version could be implemented in NinjaWEB, though this would not catch e.g. as stall in the FPGA.

In NinjaWEB, make sure that in NN24SystemClass.py the following is set:

```
self.v_src_boost = True
```

This will increase the source card supply voltage from 5.2V to 9.7V.

A similar setting is in the Matlab initStat() function: stat.v_src_boost = true;

NinjaNIRS 2027

NN27 is a further planned upgrade of the two whole-head, high-density systems at BU with **SiPM based high-detectivity optodes**, as designed and tested for the UG3 grant milestones.

Detector Optodes

Detector optodes can be manufactured by NIRx in exactly the same way as source optodes. PCB design files are in the Github repo.

Detector Splitter Boards

Eric has designed splitter boards that contain the necessary bias voltage generator. A set of these will have to be ordered, maybe including minor revisions.

Protection of the user from the ~60V bias voltage should also be considered. Potting or conformal coating could be explored.

Detector Interface Cards

The detector interface cards stay largely the same. The only change is to route 12V directly to the splitter board to power the bias generator. This can either be done by manually soldering a jumper wire on a current board, bypassing U13, or by manufacturing a new set of v3 cards with the following changes:

Designator	NN24-26	NN27	Notes
R71	DNP	0 Ohm	New component; does not exist in prev.
U13	ADP7142	DNP	
C104	10uF	DNP	
LF1		DNP	

Additional components could be DNP'd in NN27 if desired.

Backplane

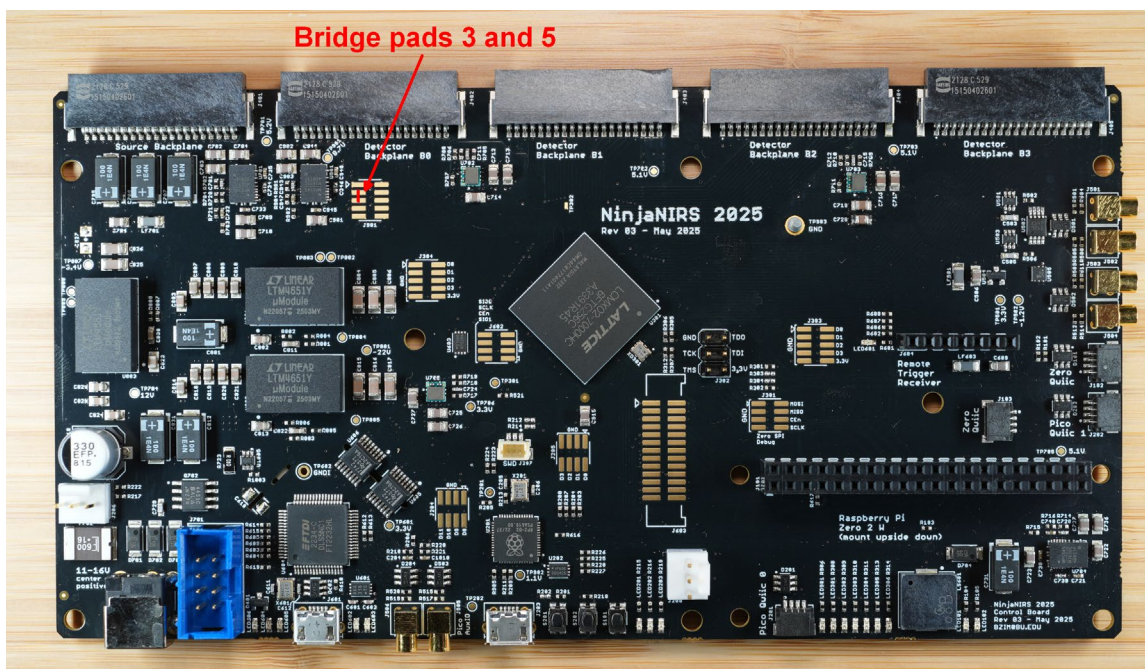
The existing backplanes are compatible.

Control Card

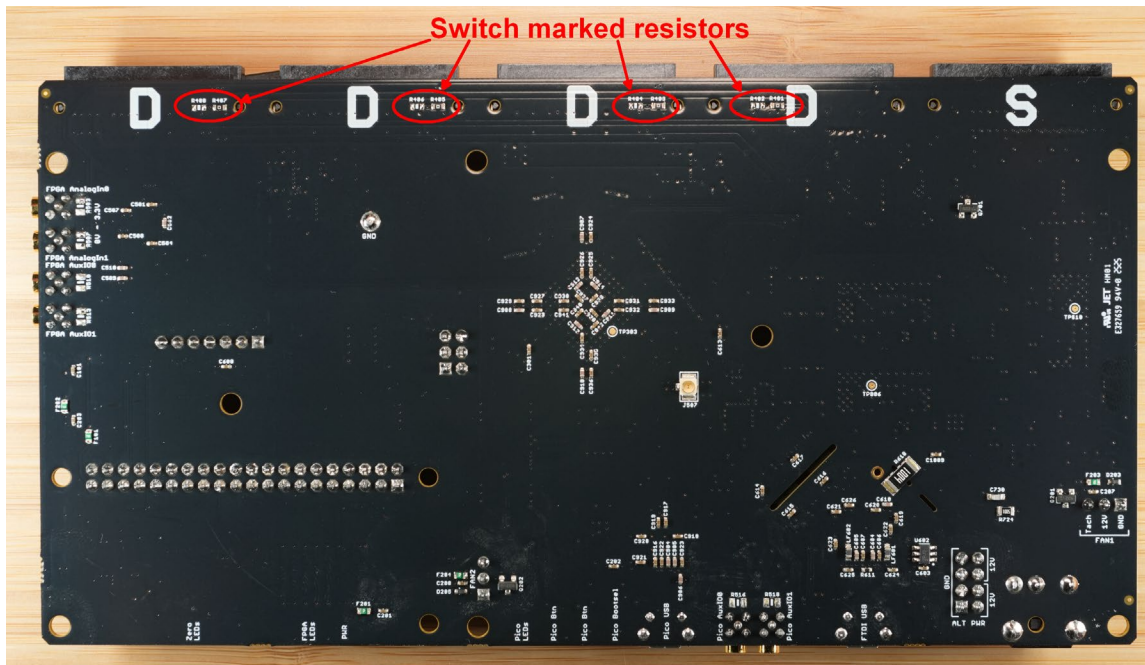
Similar to the detector cards, 12V must be routed to the backplane to ultimately power the bias generators. The v3 Control Cards already have provisions for this. V1 and V2 cards are not compatible.

Here are the necessary changes on the v3 control card:

On the front side of the card, bridge pads 3 and 5 of connector footprint J901.



On the back side of the card, move the 0 Ohm resistor from R401 to R402, from R403 to R404, from R405 to R406, and from R407 to R408.



System Case

The system case must be redesigned / enlarged to accommodate the larger detector splitter boards, and possibly to protect the user from the bias voltage.

Firmware / NinjaWeb GUI

As a starting point, no firmware needs to be changed.

However, if control of bias voltage is necessary, it might be desirable to coordinate changes with the larger system, so communication between the splitter board and detector interface board has to be implemented, and then enable communication of this new data to NinjaWeb will also require changes to the FPGA firmware.